

NEWTON MURIMI IRUNGU

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AI ENGINEER | DATA ANALYST | MACHINE LEARNING & GENERATIVE AI

PROFESSIONAL SUMMARY

AI Engineer and Data Analyst with a strong foundation in mathematics, business, machine learning, and deep learning. Experienced in building and evaluating practical AI models with Python, PyTorch, TensorFlow, Keras, and Scikit-learn, with hands-on work spanning computer vision, NLP, Generative AI, Transformers, and RAG. My Mathematics and Business Studies background gives me a practical, analytical approach to turning data and technical ideas into useful solutions.

TECHNICAL SKILLS

Programming & Data	Python, SQL, Pandas, NumPy, Data Analysis, Statistics, Data Visualization
Machine Learning	Scikit-learn, supervised learning, classification, regression, model evaluation, feature engineering
Deep Learning	PyTorch, TensorFlow, Keras, neural networks, CNNs, transfer learning
Generative AI & NLP	LLMs, NLP, embeddings, Transformers, RAG, LangChain, Hugging Face, prompt engineering
Computer Vision	Image classification, Vision Transformers (ViT), CNN-ViT hybrid architectures, image augmentation
Tools & Engineering	Git, GitHub, Jupyter, VS Code, REST APIs, FastAPI, Docker, GitHub Actions
Mathematics & Business	Linear Algebra, Multivariable Calculus, Probability & Statistics, Financial Accounting, Managerial Accounting, Business Strategy

PROFESSIONAL EXPERIENCE

AI Engineering & Data Analysis — Independent Projects

Nairobi, Kenya

Developing and documenting practical machine learning, deep learning, computer vision, NLP, and Generative AI projects. Building and evaluating models with PyTorch, TensorFlow, Keras, and Scikit-learn; working with CNNs, transfer learning, Vision Transformers, Transformers, embeddings, LLMs, and RAG; and maintaining a structured GitHub portfolio.

SELECTED AI PROJECTS

IBM AI Engineering Portfolio | Python, PyTorch, TensorFlow, Keras, Scikit-learn

Completed practical projects covering rainfall prediction, waste classification, Fashion-MNIST, neural-network prediction, data loading and augmentation, Vision Transformers, and CNN-ViT integration.

Generative AI & NLP Learning Projects | PyTorch, Transformers, Hugging Face, NLP, LLMs

Developed practical knowledge across tokenization, embeddings, Word2Vec, N-Gram language models, sequence-to-sequence RNNs, attention, masking, Transformer architectures, fine-tuning, and RAG.

PROFESSIONAL CERTIFICATIONS

IBM AI Engineering Professional Certificate — IBM / Coursera — Credly verified

Python for Data Science & Machine Learning — Kaggle / Coursera

EDUCATION

Bachelor of Education (Arts) — Mathematics & Business Studies | 2022–2026

Pioneer International University, Nairobi, Kenya

Completed coursework and practical requirements; expected graduation: November 28, 2026.

Relevant foundation: Linear Algebra, Vector Analysis, Multivariable Calculus, Probability & Statistics, Financial Accounting, Cost & Managerial Accounting, and Business Strategy.

PROFESSIONAL PORTFOLIO

GitHub: <https://github.com/newtonmurimi19-create/IBM-AI-Engineering-Professional-Certificate.git>

LinkedIn: <https://www.linkedin.com/in/newtonmurimi-720211421/>

Credly: <https://www.credly.com/users/newton-murimi>